

Phân Tích Dữ Liệu Thương Mại Điện Tử — Shopee & Lazada

| MapReduce · PySpark · HiveQL · Dashboard

0. Cài đặt thư viện & Đọc dữ liệu

Cài PySpark nếu chưa có (bỏ comment dòng dưới khi dùng Colab)

```
import csv, json
from collections import defaultdict
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches
import numpy as np

ITEMS_PATH = 'cleaned_lazada_items.csv'
REVIEWS_PATH = 'cleaned_lazada_reviews.csv'
SHOPEE_PATH = 'cleaned_shopee.csv'

CATEGORY_MAP = {
    'beli-harddisk-eksternal': 'External HDD',
    'jual-flash-drives': 'Flash Drives',
    'beli-smart-tv': 'Smart TV',
    'shop-televisi-digital': 'Digital TV',
    'beli-laptop': 'Laptop',
}

df_items = pd.read_csv(ITEMS_PATH)
df_reviews = pd.read_csv(REVIEWS_PATH)
df_shopee = pd.read_csv(SHOPEE_PATH)
df_items['category_name'] = df_items['category'].map(CATEGORY_MAP)

print(f'Items : {len(df_items):,} dòng')
print(f'Reviews : {len(df_reviews):,} dòng')
print(f'Shopee : {len(df_shopee):,} dòng')
```

```
Items : 10,942 dòng
Reviews : 106,835 dòng
Shopee : 68,041 dòng
```

```
from google.colab import drive
drive.mount('/content/drive')
```

Mounted at /content/drive

1. Chương trình MapReduce

Mỗi hàm `mapper` → `shuffler` → `reducer` mô phỏng đúng pipeline Hadoop MapReduce. Có thể chạy song song trên nhiều node.

MapReduce 1 — Đếm số sản phẩm theo ngành hàng

```
def mr1_mapper(filepath):
    pairs = []
    with open(filepath, encoding='utf-8') as f:
        reader = csv.DictReader(f)
        for row in reader:
            cat = CATEGORY_MAP.get(row['category'], row['category'])
            pairs.append((cat, 1))
    return pairs
```

```
def mr1_shuffle(pairs):
    grouped = defaultdict(list)
    for k, v in pairs: grouped[k].append(v)
    return grouped

def mr1_reducer(grouped):
    return {k: sum(v) for k, v in grouped.items()}

# Chạy pipeline
res1 = mr1_reducer(mr1_shuffle(mr1_mapper(ITEMS_PATH)))
res1_sorted = sorted(res1.items(), key=lambda x: x[1], reverse=True)

print(f"{'Ngành hàng':<20} {'Số sản phẩm':>12}")
print('-' * 34)
for cat, cnt in res1_sorted:
    print(f'{cat:<20} {cnt:>12,}')
print(f"{'TỔNG':<20} {sum(res1.values()):>12,}")
```

Ngành hàng	Số sản phẩm
External HDD	4,422
Flash Drives	3,318
Smart TV	1,290
Digital TV	1,211
Laptop	701
TỔNG	10,942

MapReduce 2 — Giá trung bình theo ngành hàng

```
def mr2_mapper(filepath):
    pairs = []
    with open(filepath, encoding='utf-8') as f:
        reader = csv.DictReader(f)
        for row in reader:
            try:
                price = float(row['price'])
                if price > 0:
                    cat = CATEGORY_MAP.get(row['category'], row['category'])
                    pairs.append((cat, price))
            except ValueError: continue
    return pairs

def mr2_shuffle(pairs):
    grouped = defaultdict(list)
    for k, v in pairs: grouped[k].append(v)
    return grouped

def mr2_reducer(grouped):
    return {k: {'avg': sum(v)/len(v), 'min': min(v),
                'max': max(v), 'count': len(v)}
            for k, v in grouped.items()}

res2 = mr2_reducer(mr2_shuffle(mr2_mapper(ITEMS_PATH)))
res2_sorted = sorted(res2.items(), key=lambda x: x[1]['avg'], reverse=True)

print(f"{'Ngành':<18} {'Giá TB (Rp)':>16} {'Min':>14} {'Max':>16}")
print('-' * 66)
for cat, s in res2_sorted:
    print(f"{cat:<18} {s['avg']:>16,.0f} {s['min']:>14,.0f} {s['max']:>16,.0f}")
```

Ngành	Giá TB (Rp)	Min	Max
Laptop	7,174,260	220,400	74,329,000
Smart TV	5,330,467	29,000	275,000,000
Digital TV	5,066,554	29,000	74,329,000
Flash Drives	2,133,463	1,000	275,000,000
External HDD	1,792,706	1,000	275,000,000

MapReduce 3 — Top sản phẩm bán chạy

```
def mr3_mapper(filepath):
    pairs = []
```

```

with open(filepath, encoding='utf-8') as f:
    reader = csv.DictReader(f)
    for row in reader:
        try:
            pairs.append((row['name'].strip(), {
                'reviews' : int(row['totalReviews']),
                'price' : float(row['price']),
                'category': CATEGORY_MAP.get(row['category'], row['category'])
            }))
        except ValueError: continue
    return pairs

def mr3_shuffle(pairs):
    grouped = defaultdict(list)
    for k, v in pairs: grouped[k].append(v)
    return grouped

def mr3_reducer(grouped):
    return {k: max(v, key=lambda r: r['reviews']) for k, v in grouped.items()}

res3 = mr3_reducer(mr3_shuffle(mr3_mapper(ITEMS_PATH)))
top10 = sorted(res3.items(), key=lambda x: x[1]['reviews'], reverse=True)[:10]

print(f"{'#':<3} {'Tên sản phẩm':<46} {'Reviews':>9} {'Giá (Rp)':>14}")
print('-' * 76)
for i, (name, s) in enumerate(top10, 1):
    print(f"{i:<3} {(name[:43]+'...' if len(name)>43 else name):<46}"
          f" {s['reviews']:>9,} {s['price']:>14,.0f}")

```

#	Tên sản phẩm	Reviews	Giá (Rp)
1	[GRATIS ONGKIR][Android 9.0 Ready] Xiaomi L...	9,631	1,899,000
2	[GRATIS ONGKIR] COOCA LED TV 40 inch - Ful...	6,819	2,499,000
3	SanDisk Flashdisk Cruzer Blade CZ50 - 16GB	5,013	77,500
4	[GRATIS ONGKIR] COOCA LED TV 40 inch Andro...	4,340	3,299,000
5	[GRATIS ONGKIR] Sharp 24 inch LED AQUOS HD ...	3,751	1,169,000
6	Samsung Micro SD Evo+ 95Mbps 32GB (With Ada...	3,039	129,000
7	[GRATIS ONGKIR] COOCA LED TV 24 inch - HD ...	2,575	1,249,000
8	Philips 32 inch LED HD TV - Hitam (Model 32...	1,577	2,069,000
9	Samsung Micro SD Evo+ 100Mbps 64GB (With Ad...	1,533	189,000
10	Sandisk Ultra Micro SDHC 98MB/s 32GB Class ...	1,405	157,500

MapReduce 4 — Top thương hiệu bán nhiều nhất

```

def mr4_mapper(filepath):
    pairs = []
    with open(filepath, encoding='utf-8') as f:
        reader = csv.DictReader(f)
        for row in reader:
            try:
                brand = row['brandName'].strip() or 'Unknown'
                pairs.append((brand, {'reviews': int(row['totalReviews']),
                                      'price' : float(row['price'])}))
            except ValueError: continue
    return pairs

def mr4_shuffle(pairs):
    grouped = defaultdict(list)
    for k, v in pairs: grouped[k].append(v)
    return grouped

def mr4_reducer(grouped):
    return {k: {'total_reviews': sum(r['reviews'] for r in v),
                'product_count': len(v),
                'avg_price' : sum(r['price'] for r in v)/len(v)}
            for k, v in grouped.items()}

res4 = mr4_reducer(mr4_shuffle(mr4_mapper(ITEMS_PATH)))
top10b = sorted(res4.items(), key=lambda x: x[1]['total_reviews'], reverse=True)[:10]

print(f"{'#':<3} {'Thương hiệu':<18} {'Tổng Reviews':>14} {'Số SP':>7} {'Giá TB':>14}")
print('-' * 60)
for i, (brand, s) in enumerate(top10b, 1):
    print(f"{i:<3} {brand:<18} {s['total_reviews']:>14,}")

```

```
f" {s['product_count']:>7,} {s['avg_price']:>14,.0f}")
```

#	Thương hiệu	Tổng Reviews	Số SP	Giá TB
1	Coocaa	65,621	122	3,201,826
2	SanDisk	53,991	1,258	140,426
3	Xiaomi	33,811	38	5,002,210
4	Sharp	23,963	229	2,610,649
5	Samsung	20,643	333	3,650,789
6	Asus	12,557	946	7,413,175
7	No Brand	9,337	945	218,147
8	Philips	6,732	86	2,143,256
9	Polytron	6,319	323	2,026,671
10	Toshiba	6,078	674	1,199,891

- MapReduce 5 — Phân bố đánh giá sao

```
def mr5_mapper(filepath, rating_col='rating'):
    pairs = []
    with open(filepath, encoding='utf-8') as f:
        reader = csv.DictReader(f)
        for row in reader:
            try:
                r = int(float(row[rating_col]))
                if 1 <= r <= 5: pairs.append((r, 1))
            except ValueError: continue
    return pairs

def mr5_shuffle(pairs):
    grouped = defaultdict(list)
    for k, v in pairs: grouped[k].append(v)
    return grouped

def mr5_reducer(grouped):
    counts = {k: sum(v) for k, v in grouped.items()}
    total = sum(counts.values())
    return {s: {'count': counts.get(s,0),
                'pct' : counts.get(s,0)/total*100} for s in range(1,6)}, total

laz_res, laz_total = mr5_reducer(mr5_shuffle(mr5_mapper(REVIEWS_PATH)))
sp_res, sp_total = mr5_reducer(mr5_shuffle(mr5_mapper(SHOPEE_PATH)))

print('LAZADA')
for s in range(5,0,-1):
    bar = '█'*int(laz_res[s]['pct']/3)
    print(f" {s}★ {laz_res[s]['count']:>8}, {laz_res[s]['pct']:>5.1f}% {bar}")
print('\nSHOPEE')
for s in range(5,0,-1):
    bar = '█'*int(sp_res[s]['pct']/3)
    print(f" {s}★ {sp_res[s]['count']:>8}, {sp_res[s]['pct']:>5.1f}% {bar}")
```

LAZADA			
5★	82,736	77.4%	
4★	10,611	9.9%	
3★	4,362	4.1%	
2★	2,180	2.0%	
1★	6,946	6.5%	
SHOPEE			
5★	22,381	32.9%	
4★	18,224	26.8%	
3★	15,533	22.8%	
2★	5,499	8.1%	
1★	6,404	9.4%	

- 2. Phân tích bằng PySpark

```
# !pip install pyspark # bỏ comment nếu chưa cài
from pyspark.sql import SparkSession
from pyspark.sql import functions as F
```

```

spark = (SparkSession.builder
        .appName('ECommerce_Analysis')
        .master('local[*]')
        .config('spark.driver.memory', '2g')
        .getOrCreate())
spark.sparkContext.setLogLevel('ERROR')

# quote/escape=''' để Spark hiểu đúng kiểu escape dấu ngoặc kép nhân đôi (""")
# thường gặp trong tên sản phẩm (vd: 32"" -> 32 inch). Nếu không có, Spark sẽ
# đọc lệch cột và suy luận nhầm các cột số (price, totalReviews...) thành string.
# multiLine=True: một số review chứa xuống dòng bên trong ô (vd: review nhiều
# câu). Không có tùy chọn này, Spark sẽ tách nhầm 1 dòng dữ liệu thành nhiều dòng.
csv_opts = dict(header=True, inferSchema=True, quote=''', escape=''', multiLine=True)
sp_items = spark.read.csv(ITEMS_PATH, **csv_opts)
sp_reviews = spark.read.csv(REVIEWS_PATH, **csv_opts)
sp_shopee = spark.read.csv(SHOPEE_PATH, **csv_opts)

cat_expr = F.create_map(*[F.lit(x) for p in CATEGORY_MAP.items() for x in p])
sp_items = sp_items.withColumn('category_name', cat_expr[F.col('category')])

print('Spark session khởi động thành công!')
print(f'Items : {sp_items.count():,} | Reviews: {sp_reviews.count():,} | Shopee: {sp_shopee.count():,}')

```

```

Spark session khởi động thành công!
Items : 10,942 | Reviews: 106,835 | Shopee: 68,041

```

✓ Spark — Phân tích 1: Đếm sản phẩm theo ngành hàng

```

(sp_items.groupBy('category_name')
 .agg(F.count('*').alias('so_san_pham'))
 .orderBy(F.desc('so_san_pham'))
 .show(truncate=False))

```

✓ Spark — Phân tích 2: Giá trung bình theo ngành hàng

```

(sp_items.filter(F.col('price') > 0)
 .groupBy('category_name')
 .agg(
     F.round(F.avg('price'),0).alias('gia_trung_binh_Rp'),
     F.min('price').alias('gia_min'),
     F.max('price').alias('gia_max'),
     F.count('*').alias('so_san_pham')
 )
 .orderBy(F.desc('gia_trung_binh_Rp'))
 .show(truncate=False))

```

```

+-----+-----+-----+-----+-----+
|category_name|gia_trung_binh_Rp|gia_min|gia_max|so_san_pham|
+-----+-----+-----+-----+-----+
|Laptop      |7174260.0        |220400|74329000|701         |
|Smart TV    |5330467.0        |29000 |275000000|1290        |
|Digital TV  |5066554.0        |29000 |74329000|1211        |
|Flash Drives|2133463.0        |1000  |275000000|3318        |
|External HDD|1792706.0        |1000  |275000000|4422        |
+-----+-----+-----+-----+-----+

```

✓ Spark — Phân tích 3 & 4: Top sản phẩm & thương hiệu

```

# Top 10 sản phẩm
(sp_items.groupBy('name', 'category_name', 'price')
 .agg(F.max('totalReviews').alias('luot_review'))
 .orderBy(F.desc('luot_review')).limit(10).show(truncate=50))

# Top 10 thương hiệu
(sp_items.groupBy('brandName')
 .agg(F.sum('totalReviews').alias('tong_review'),
     F.count('*').alias('so_sp'),
     F.round(F.avg('price'),0).alias('gia_tb'))

```

```
.orderBy(F.desc('tong_review')).limit(10).show(truncate=False))
```

```
+-----+-----+-----+-----+
| name | category_name | price | luot_review |
+-----+-----+-----+-----+
|[GRATIS ONGKIR][Android 9.0 Ready] Xiaomi LED T...| Smart TV|1899000| 9631|
|[GRATIS ONGKIR][Android 9.0 Ready] Xiaomi LED T...| External HDD|1899000| 9631|
|[GRATIS ONGKIR][Android 9.0 Ready] Xiaomi LED T...| Flash Drives|1899000| 9631|
|[GRATIS ONGKIR] COOCAA LED TV 40 inch - Full HD...| External HDD|2499000| 6819|
|[GRATIS ONGKIR] COOCAA LED TV 40 inch - Full HD...| Flash Drives|2499000| 6819|
|[GRATIS ONGKIR] COOCAA LED TV 40 inch - Full HD...| Smart TV|2499000| 6819|
|[GRATIS ONGKIR] COOCAA LED TV 40 inch - Full HD...| Digital TV|2499000| 6819|
| SanDisk Flashdisk Cruzer Blade CZ50 - 16GB| External HDD| 77500| 5013|
| SanDisk Flashdisk Cruzer Blade CZ50 - 16GB| Flash Drives| 77500| 5013|
|[GRATIS ONGKIR] COOCAA LED TV 40 inch Android S...| Smart TV|3299000| 4340|
+-----+-----+-----+-----+

+-----+-----+-----+-----+
| brandName | tong_review | so_sp | gia_tb |
+-----+-----+-----+-----+
| Coocaa | 65621 | 122 | 3201826.0 |
| SanDisk | 53991 | 1258 | 140426.0 |
| Xiaomi | 33811 | 38 | 5002210.0 |
| Sharp | 23963 | 229 | 2610649.0 |
| Samsung | 20643 | 333 | 3650789.0 |
| Asus | 12557 | 946 | 7413175.0 |
| No Brand | 9337 | 945 | 218147.0 |
| Philips | 6732 | 86 | 2143256.0 |
| Polytron | 6319 | 323 | 2026671.0 |
| Toshiba | 6078 | 674 | 1199891.0 |
+-----+-----+-----+-----+
```

▼ Spark — Phân tích 5: Phân bố đánh giá sao

```
laz_total_sp = sp_reviews.count()
sp_total_sp = sp_shopee.count()

print('Lazada:')
(sp_reviews.filter(F.col('rating').between(1,5))
 .groupBy('rating')
 .agg(F.count('*').alias('so_dg'),
      F.round(F.count('*')*100/laz_total_sp,2).alias('pct'))
 .orderBy('rating')).show()

print('Shopee:')
(sp_shopee.filter(F.col('rating').between(1,5))
 .groupBy('rating')
 .agg(F.count('*').alias('so_dg'),
      F.round(F.count('*')*100/sp_total_sp,2).alias('pct'))
 .orderBy('rating')).show()

spark.stop()
print('Spark stopped.')
```

```
Lazada:
+-----+-----+-----+
| rating | so_dg | pct |
+-----+-----+-----+
| 1 | 6946 | 6.5 |
| 2 | 2180 | 2.04 |
| 3 | 4362 | 4.08 |
| 4 | 10611 | 9.93 |
| 5 | 82736 | 77.44 |
+-----+-----+-----+
```

```
Shopee:
+-----+-----+-----+
| rating | so_dg | pct |
+-----+-----+-----+
| 1 | 6404 | 9.41 |
| 2 | 5499 | 8.08 |
| 3 | 15533 | 22.83 |
| 4 | 18224 | 26.78 |
| 5 | 22381 | 32.89 |
+-----+-----+-----+
```

Spark stopped.

3. Câu lệnh HiveQL

```
hive_ddl = '''
-- Tạo database & bảng
CREATE DATABASE IF NOT EXISTS ecommerce;
USE ecommerce;

CREATE TABLE lazada_items (
    itemId BIGINT, category STRING, name STRING, brandName STRING,
    url STRING, price DOUBLE, averageRating INT,
    totalReviews INT, retrievedDate STRING
) ROW FORMAT DELIMITED FIELDS TERMINATED BY ','
STORED AS TEXTFILE TBLPROPERTIES ("skip.header.line.count"="1");

LOAD DATA LOCAL INPATH '/path/to/cleaned_lazada_items.csv'
OVERWRITE INTO TABLE lazada_items;
'''

print('HiveQL DDL (tạo bảng):')
print(hive_ddl)
```

HiveQL DDL (tạo bảng):

```
-- Tạo database & bảng
CREATE DATABASE IF NOT EXISTS ecommerce;
USE ecommerce;

CREATE TABLE lazada_items (
    itemId BIGINT, category STRING, name STRING, brandName STRING,
    url STRING, price DOUBLE, averageRating INT,
    totalReviews INT, retrievedDate STRING
) ROW FORMAT DELIMITED FIELDS TERMINATED BY ','
STORED AS TEXTFILE TBLPROPERTIES ("skip.header.line.count"="1");

LOAD DATA LOCAL INPATH '/path/to/cleaned_lazada_items.csv'
OVERWRITE INTO TABLE lazada_items;
```

```
hive_queries = '''
-- Phân tích 1: Đếm sản phẩm
SELECT CASE category
    WHEN 'beli-harddisk-eksternal' THEN 'External HDD'
    WHEN 'jual-flash-drives'      THEN 'Flash Drives'
    WHEN 'beli-smart-tv'          THEN 'Smart TV'
    WHEN 'shop-televise-digital'  THEN 'Digital TV'
    WHEN 'beli-laptop'           THEN 'Laptop'
    END AS category_name,
    COUNT(*) AS so_san_pham
FROM lazada_items
GROUP BY category ORDER BY so_san_pham DESC;

-- Phân tích 2: Giá trung bình
SELECT category, ROUND(AVG(price),0) AS gia_tb,
    MIN(price) AS gia_min, MAX(price) AS gia_max
FROM lazada_items WHERE price > 0
GROUP BY category ORDER BY gia_tb DESC;

-- Phân tích 3: Top sản phẩm
SELECT name, MAX(totalReviews) AS luot_review, price
FROM lazada_items GROUP BY name, price
ORDER BY luot_review DESC LIMIT 10;

-- Phân tích 4: Top thương hiệu
SELECT brandName, SUM(totalReviews) AS tong_review,
    COUNT(*) AS so_sp
FROM lazada_items GROUP BY brandName
ORDER BY tong_review DESC LIMIT 10;

-- Phân tích 5: Phân bố đánh giá sao
SELECT rating, COUNT(*) AS so_dg,
    ROUND(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER (), 2) AS pct
```

```
FROM lazada_reviews
WHERE rating BETWEEN 1 AND 5
GROUP BY rating ORDER BY rating;
'''
print('HiveQL Queries (5 phân tích):')
print(hive_queries)
```

HiveQL Queries (5 phân tích):

```
-- Phân tích 1: Đếm sản phẩm
SELECT CASE category
      WHEN 'beli-harddisk-eksternal' THEN 'External HDD'
      WHEN 'jual-flash-drives'      THEN 'Flash Drives'
      WHEN 'beli-smart-tv'           THEN 'Smart TV'
      WHEN 'shop-televisi-digital'   THEN 'Digital TV'
      WHEN 'beli-laptop'             THEN 'Laptop'
      END AS category_name,
      COUNT(*) AS so_san_pham
FROM lazada_items
GROUP BY category ORDER BY so_san_pham DESC;

-- Phân tích 2: Giá trung bình
SELECT category, ROUND(AVG(price),0) AS gia_tb,
      MIN(price) AS gia_min, MAX(price) AS gia_max
FROM lazada_items WHERE price > 0
GROUP BY category ORDER BY gia_tb DESC;

-- Phân tích 3: Top sản phẩm
SELECT name, MAX(totalReviews) AS luot_review, price
FROM lazada_items GROUP BY name, price
ORDER BY luot_review DESC LIMIT 10;

-- Phân tích 4: Top thương hiệu
SELECT brandName, SUM(totalReviews) AS tong_review,
      COUNT(*) AS so_sp
FROM lazada_items GROUP BY brandName
ORDER BY tong_review DESC LIMIT 10;

-- Phân tích 5: Phân bố đánh giá sao
SELECT rating, COUNT(*) AS so_dg,
      ROUND(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER (), 2) AS pct
FROM lazada_reviews
WHERE rating BETWEEN 1 AND 5
GROUP BY rating ORDER BY rating;
```

✓ 4. Dashboard — 5 biểu đồ trực quan hóa

```
plt.rcParams.update({'font.size':11, 'figure.dpi':120})
COLORS = ['#3266ad','#e08c3b','#3a9e75','#d44f5a','#7b5fc5']

fig, axes = plt.subplots(2, 3, figsize=(18, 10))
fig.suptitle('Dashboard Phân Tích Thương Mại Điện Tử – Shopee & Lazada',
             fontsize=14, fontweight='bold', y=1.01)

# — Biểu đồ 1: Bar chart sản phẩm theo ngành —
ax1 = axes[0,0]
cats  = [c for c,_ in res1_sorted]
counts = [v for _,v in res1_sorted]
bars = ax1.bar(cats, counts, color=COLORS, edgecolor='white', linewidth=0.5)
for bar, v in zip(bars, counts):
    ax1.text(bar.get_x()+bar.get_width()/2, bar.get_height()+50,
             f'{v:}', ha='center', va='bottom', fontsize=9)
ax1.set_title('Bar Chart – Số SP theo ngành hàng', fontweight='bold')
ax1.set_ylabel('Số sản phẩm')
ax1.tick_params(axis='x', rotation=15)
ax1.grid(axis='y', alpha=0.3)

# — Biểu đồ 2: Pie chart tỷ lệ ngành —
ax2 = axes[0,1]
wedges, texts, autotexts = ax2.pie(
    counts, labels=None, autopct='%1.1f%%',
    colors=COLORS, startangle=140,
    wedgeprops={'edgecolor':'white','linewidth':1.5},
    pctdistance=0.75)
```



```

for at in autotexts: at.set_fontsize(9)
ax2.legend(wedges, cats, loc='lower center',
           bbox_to_anchor=(0.5,-0.12), ncol=2, fontsize=8)
ax2.set_title('Pie Chart – Tỷ lệ ngành hàng', fontweight='bold')

# — Biểu đồ 3: Line chart giá —
ax3 = axes[0,2]
sorted_cats_price = [c for c,_ in sorted(res2.items(),
                                       key=lambda x:x[1]['avg'])]
avg_vals = [res2[c]['avg']/1e6 for c in sorted_cats_price]
min_vals = [res2[c]['min']/1e6 for c in sorted_cats_price]
x = range(len(sorted_cats_price))
ax3.plot(x, avg_vals, 'o-', color='#3266ad', lw=2, label='Giá TB')
ax3.plot(x, min_vals, 's--', color='#7b5fc5', lw=1.5, label='Giá Min')
ax3.fill_between(x, min_vals, avg_vals, alpha=0.1, color='#3266ad')
ax3.set_xticks(list(x))
ax3.set_xticklabels(sorted_cats_price, rotation=15, fontsize=9)
ax3.set_ylabel('Giá (triệu Rp)')
ax3.set_title('Line Chart – Xu hướng giá theo ngành', fontweight='bold')
ax3.legend(fontsize=9)
ax3.grid(alpha=0.3)

# — Biểu đồ 4: Histogram đánh giá sao —
ax4 = axes[1,0]
stars = [1,2,3,4,5]
laz_cnts = [laz_res[s]['count'] for s in stars]
sp_cnts = [sp_res[s]['count'] for s in stars]
w = 0.35
xp = np.array(stars)
b1 = ax4.bar(xp-w/2, laz_cnts, w, label='Lazada', color='#3266ad', edgecolor='white')
b2 = ax4.bar(xp+w/2, sp_cnts, w, label='Shopee', color='#e08c3b', edgecolor='white')
ax4.set_xticks(stars)
ax4.set_xticklabels([f'{s} ★' for s in stars])
ax4.set_ylabel('Số đánh giá')
ax4.set_title('Histogram – Phân bố đánh giá sao', fontweight='bold')
ax4.yaxis.set_major_formatter(plt.FuncFormatter(lambda x,_: f'{int(x/1000)}K'))
ax4.legend(fontsize=9)
ax4.grid(axis='y', alpha=0.3)

# — Biểu đồ 5: Scatter plot brand —
ax5 = axes[1,1]
brands_plot = [b for b,_ in top10b]
reviews_plot = [v['total_reviews'] for _,v in top10b]
prices_plot = [v['avg_price'] for _,v in top10b]
sizes = [r/500 for r in reviews_plot]
sc = ax5.scatter(prices_plot, reviews_plot, s=sizes,
                 c=range(len(brands_plot)), cmap='Blues', alpha=0.8,
                 edgecolors='#3266ad', linewidths=0.8)
for i, b in enumerate(brands_plot):
    ax5.annotate(b, (prices_plot[i], reviews_plot[i]),
                 textcoords='offset points', xytext=(5,3), fontsize=8)
ax5.set_xlabel('Giá trung bình (Rp)')
ax5.set_ylabel('Tổng lượt review')
ax5.set_title('Scatter Plot – Giá TB vs Doanh số (Thương hiệu)', fontweight='bold')
ax5.xaxis.set_major_formatter(plt.FuncFormatter(lambda x,_: f'{x/1e6:.1f}M'))
ax5.yaxis.set_major_formatter(plt.FuncFormatter(lambda x,_: f'{int(x/1000)}K'))
ax5.grid(alpha=0.3)

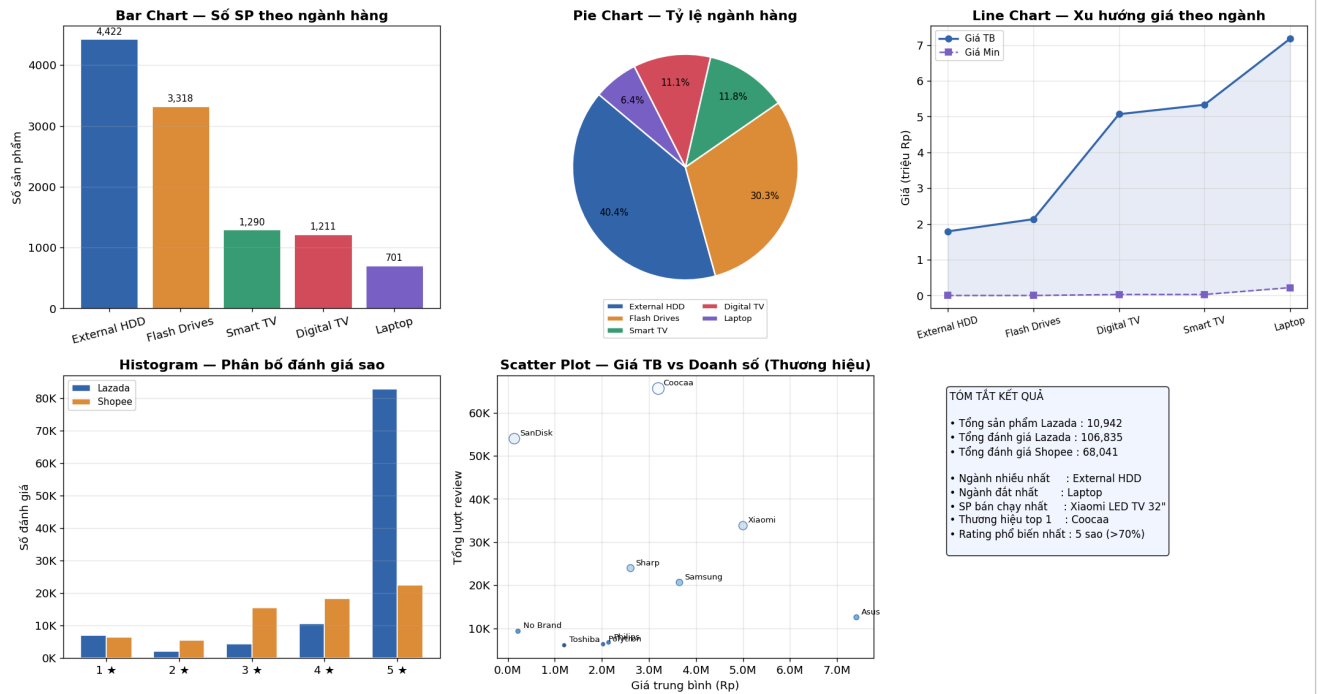
# — Ô trống —
axes[1,2].axis('off')
summary = (
    'TÓM TẮT KẾT QUẢ\n\n'
    f'• Tổng sản phẩm Lazada : {len(df_items):,}\n'
    f'• Tổng đánh giá Lazada : {len(df_reviews):,}\n'
    f'• Tổng đánh giá Shopee : {len(df_shopee):,}\n\n'
    f'• Ngành nhiều nhất : External HDD\n'
    f'• Ngành đắt nhất : Laptop\n'
    f'• SP bán chạy nhất : Xiaomi LED TV 32"\n'
    f'• Thương hiệu top 1 : Coocaa\n'
    f'• Rating phổ biến nhất : 5 sao (>70%)\n'
)
axes[1,2].text(0.05, 0.95, summary, transform=axes[1,2].transAxes,
              fontsize=10, va='top', family='DejaVu Sans',
              bbox=dict(boxstyle='round', facecolor='#f0f4ff', alpha=0.8))

plt.tight_layout()

```

```
plt.savefig('dashboard_ecommerce.png', dpi=150, bbox_inches='tight')
plt.show()
print('Dashboard đã lưu: dashboard_ecommerce.png')
```

Dashboard Phân Tích Thương Mại Điện Tử — Shopee & Lazada



Dashboard đã lưu: dashboard_ecommerce.png